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Saturn's Zonal Winds at Cloud Level

Agustín Sanchez-Lavega and José F. Rojas

Departamento de Física Aplicada I, E.T.S. Ingenieros—Universidad Pais Vasco, Alda. Urquijo s/n, 48013 Bilbao, Spain
E-mail: wupsalaa@bi.ehu.es

and

Pedro V. Sada

Departamento de Física y Mathemáticas, Universidad de Monterrey, Av. I. Morones Prieto 4500 Pte., Garza García N. L. 66238 México

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We present an analysis of new cloud motion measurements in Saturn's atmosphere from *Voyager 1* and *2* images and define a nominal zonal wind velocity vs latitude profile for the cloud top level that can be used as reference for the upcoming *Cassini* mission. Improvements over previous studies include: (i) twice the number of wind vectors used, (ii) independent measurements of cloud tracers, (iii) extended latitudinal coverage, (iv) strict error control and evaluation, and (v) a new zonal averaging methodology. Mean zonal wind velocities are given for the latitude range 81° to −71° (planetographic) in steps of 0.5° with a gap between 0° and −10° due to ring shadowing. This research also includes the study of the morphology and motions of cloud systems in the equator and in the southern hemisphere which have not been described previously. We give an ephemeris (motion equation) for several long-lived features that could be observable if they survive until *Cassini* arrives on Saturn. The long-term behavior of the zonal wind profile is studied by comparing it with a complete set of wind velocities compiled from historical and modern ground-based and HST observations. Significant changes are noted in the equatorial jet between this set and the nominal *Voyager* profile. We discuss in detail the possible origin for such differences. © 2000 Academic Press

1. INTRODUCTION

Our understanding of the meteorology and dynamics of the atmosphere of a planet relies on the knowledge and precise measurements of the three-dimensional structure of mass motions (winds) and in their temporal changes. Motions in the atmosphere of Saturn were inferred for the first time by W. Herschel in 1793 (Alexander 1962). He measured the equatorial rotation of the planet, tracking dark spots for several days. This technique (i.e., cloud tracking of distinct features on telescopic images) was used successfully during the 19th and 20th centuries and provided a gross picture of Saturn's atmospheric differential rotation at the cloud top level (Sanchez-Lavega 1982). The arrival at Saturn of the *Voyager 1* and *2* spacecraft in 1980 and 1981 provided a new detailed picture of the cloud motions because of the 20–100 times higher spatial resolution obtained compared with ground-based telescopes (Smith *et al.* 1981, 1982; Ingersoll *et al.* 1984). Complementary temperature measurements and the use of the thermal wind equation helped to infer the vertical nature of the motions above the clouds (Conrath and Pirraglia 1983). During the 1990s the extensive use of sensitive CCD cameras (coupled to ground-based telescopes and the Hubble Space Telescope), and the development of giant storms and other cloud forms in Saturn's atmosphere, provided the possibility to study motions on a large temporal scale (Sanchez-Lavega *et al.* 1991, 1993a, 1996; Beebe *et al.* 1992; Barnet *et al.* 1992b).

The generally referenced zonal wind velocity profile for Saturn's atmosphere at the cloud top level is that published as Fig. 4 in Ingersoll *et al.* (1984). It is based on the tracking of about 1000 cloud features on high-resolution *Voyager 2* Imaging Science Subsystem (ISS) green-filtered images obtained during the encounter in August 1981. According to the authors, it represents a reworked version of that previously published by Smith *et al.* (1982). The upcoming encounter of the *Cassini* spacecraft with the planet in the year 2004 initially motivated us to reanalyze this profile independently of previous authors, with the aim to improve the accuracy and error determination of the motion measurements. Differences in zonal wind velocities measured by various teams on *Voyager 1* and *2* images are observed when comparing their published profiles (Smith *et al.* 1981, 1982; Sromovsky *et al.* 1983; Ingersoll *et al.* 1984; Beebe 1990 [published as Fig. 10 in Barnet *et al.* 1992a]). We thus analyzed, using a new approach to reduce the data, a new set of *Voyager* zonal winds containing twice the number of measured points than preceding works. These measurements were independently performed by two of us using a similar technique. An important goal of our analysis is to present a useful mean zonal wind profile in a table (including error estimations), in the same way as was done for Jupiter by Limaye (1986) for example.

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A second motivation for this work was to extend the latitudinal coverage of the Ingersoll *et al.* (1984) wind profile. This has been done in two ways: (i) by measuring vector displacements at greater northern latitudes on *Voyager 2* images, and (ii) by measuring cloud features in *Voyager 1* images which cover the southern equatorial latitudes occulted by the ring systems as seen by the *Voyager 2* cameras. Measurements of the *Voyager 1* images also provide the opportunity to look for short temporal changes during the nine-month interval elapsed between the *Voyager 1* and *2* encounters. Again, we found important differences between the published *Voyager 1* and *2* wind data (references above).

As a byproduct of this effort to establish as firmly as possible the location and magnitude of the Saturn's jet streams, we also performed a detailed study of the motions and characteristics of some particular features not analyzed by the *Voyager* imaging team. This analysis concentrated on a family of plumes in the northern equatorial zone (EZ) and on waves extending over some longitudes in the southern hemisphere. In addition we tracked the four long-lived distinct ovals observed during the *Voyager* encounters, establishing their motion equation (ephemeris) with the purpose of recovering them if they survive until the exploration of the saturnian system by the *Cassini* mission.

Saturn has important seasonal insolation changes due to the tilt of its rotation axis, significantly oblate geometry, scattering and thermal emission by the ring system, and equatorial ring shadowing. It has been suggested that this could produce measurable changes in the wind velocities (Barnet *et al.* 1992a). Thus, another objective of this research has been to look for long-term temporal changes in the wind profile. To do this we used a selected number of reliable historical ground-based measurements (up to the *Voyager* encounters), and a new set of recently acquired ground-based and HST data. Comparison of our nominal *Voyager* profile with the other velocity data shows significant differences. We discuss their possible origin in the last section.

2. THE VOYAGER IMAGE DATA

Images of Saturn were obtained with the ISS instrument onboard the *Voyager* spacecraft using wide angle (WA) and narrow angle (NA) Vidicon cameras (see Smith *et al.* 1977 for details on the cameras and filters, and the references in Table I for general discussions on the image characteristics). The *Voyager 1* encounter took place on 12 November 1980 at 23:46 UT (a distance from the planet of 126,000 km), and the *Voyager 2* encounter was on 26 August 1981 at 03:34 UT (a distance of 100,800 km). Copies of most of the images were acquired on CD-ROMs from the Planetary Data System (PDS) deep archive located at the National Space Science Data Center (NSSDC). Additional data were supplied by the PDS Imaging and Atmospheres Node where they where being processed.

2.1. Image Selection

Wind measurements were performed on images taken around the encounter dates in order to get the highest possible spatial resolution: on 5–12 November 1980 (*Voyager 1*), and between 19 August and 4 September 1981 (*Voyager 2*). Because of the encounter geometry, the northern hemisphere of the planet was observed at higher resolution (typically 50 km/pixel) than the southern hemisphere (150 km/pixel). Green-filtered images (effective wavelength of 566 nm) were extensively used because they showed the clouds with higher contrast than in other filters. Sometimes we complemented them with blue (479 nm) and orange (591 nm) images. *Voyager 2* images were exposed longer than those *Voyager 1*, so the Vidicon was better saturated, and hence the signal-to-noise ratio and dynamic range of the *Voyager 2* images are better than those of *Voyager 1*. This is why most researchers have used *Voyager 2* images for cloud morphology and motion studies. We selected and measured 13 pairs of *Voyager 1* and 34 pairs of *Voyager 2* images to determine the wind profile from a total of 2074 wind vector measurements. This represents twice the number of points measured by Ingersoll *et al.* (1984), and many times those used by other authors. In Table I we summarize the current status of Saturn wind measurements.

The image selection criteria were different for long-term measurements of particular major features (treated in Section 3c). High-resolution (NA) images were used for accurate size determination and morphology studies. However, the identification and determination of longitude drift rates for these features were done using all available images in order to increase the temporal coverage as much as possible. Typically these features were first detected ∼2 months before the encounters. Measurements were performed in any filter provided that the feature could be identified.

2.2. Cloud Tracers Visual Aspect

The cloud morphology and the size and number of the cloud elements used as tracers show important variations with latitude at visible wavelengths. There are analogies to and differences with the morphology seen in Jupiter's atmosphere. Detailed descriptions with images, and studies of the most conspicuous features, have been published by many teams (Smith *et al.* 1981, 1982; Hunt *et al.* 1982; Godfrey *et al.* 1983; Sromovsky *et al.* 1983; Ingersoll *et al.* 1984; Godfrey and Moore 1986; Godfrey 1988). We summarize their essential ideas with regard to wind studies:

(a) There are few longlived oval-shaped spots (vortices). They are mainly anticyclones with east–west sizes ranging from ∼1000 to 10,000 km. Their number was about 12 during the *Voyager 2* encounter (see Section 3c for details on the most conspicuous).

(b) Most cloud elements are basically patchy and concentrated in low-albedo bands around the westward jet peaks. These are the usual tracers (sizes ∼200–500 km). However, these

TABLE I
Voyager ISS Measurements of Saturn's Cloud Level Circulation

Author	Spacecraft and filter	Software	Latitude range	Number of points (N)	Notes
Smith *et al.* (1981)	V1 (gr, or)	AMOS, McIDAS	+60° to −55° (gap 0° to −10°)	∼250	(1)
Smith *et al.* (1982)	V2 (gr)	AMOS	+80° to −75° (gap 0° to −20°)	∼325	
Sromovsky *et al.* (1983)	V2 (gr)	McIDAS Cylindrical mosaics	+49° to +15°	786	(2)
Ingersoll *et al.* (1984)	V2 (gr)	AMOS	+70° to −65° (gap 0° to −25°)	∼1000	(3)
Godfrey (1986)	V2 (gr)	Polar projections	+87° to +65°	∼800	(4)
Beebe (1990)	V1 V2	VICAR	+80° to −70° (gap 0° to −10°)	∼125 (V2) ∼31 (V1)	(5)
This work	V1 (gr, bl) V2 (gr)	VICAR	+81° to −71° (gap −1° to −10°)	353 (V1) 1722 (V2)	(6)

Note. Wind velocities were determined from cloud tracking on image pairs typically separated by ∼10 h. V1 = Voyager 1, V2 = Voyager 2, bl = blue, gr = green, or = orange. (1) 8 image pairs (gr) + 2 pairs (or); (2) 16 image pairs; (3) 26 image pairs, reworked data from Smith *et al.* (1982); (4) N estimated from Fig. 4A, C; (5) published in Barnet *et al.* (1992a, Fig. 10); (6) 13 image pairs (V1) + 34 image pairs (V2).

clouds become elongated and fewer in the peaks of the eastward jets, so fewer data points are available at these latitudes.

(c) Active regions (convective in origin) with turbulent V-shaped patterns of clouds were extensively observed at mid-northern latitudes within a cyclonic region. Plume-like series of spots with rapid evolution were also seen in the equator (see Section 3c). They probably represent small-scale versions of giant convective storms (Section 4).

(d) Two unique planetary waves were seen in the northern hemisphere (the "ribbon" at 46°N and the "hexagon" at 76°N). We present evidence in Section 3c of two other waves in the southern hemisphere, with partial longitude extent.

Figure 1 shows a histogram of the number of tracers employed vs latitude. It is important to note, when determining the wind velocity profile, that tracers of type (c) and (d) could have a velocity component related to regional motions (type c) and to wave motions (type d). This and other aspects affecting the wind profile are discussed in detail in Section 5. Except in few latitudes (e.g., from −10° to −20°) full longitudinal coverage was in general accomplished.

2.3. Measurement Technique

Wind velocity measurements were independently performed by two of us (A.S.L. and Sada) using the same methodology: cloud tracking on frames separated by a given temporal interval. The procedure to geometrically correct and navigate these images was performed using the LAIA software package (developed by J. A. Cano from GEA, Spain) and the VICAR software. LAIA incorporates the VICAR code that is based on the New Mexico State University software version of the Jet Propulsion Laboratory's Multimission Image Processing Laboratory VICAR (Jepsen *et al.* 1980) as implemented by C. Barnet. The image reduction steps and measurements were performed on PC-DOS desktop computers. The pertinent navigation information was extracted from the Supplementary Experimental Data Record (SEDR) files.

The method used to transform frame coordinates to latitude–longitude positions (planetographic and/or planetocentric and System III rotation) follows the standard procedures used in the *Voyager* analysis (Beebe *et al.* 1980, Ingersoll *et al.* 1981). The planet center location in the field of view was done in either of the two following forms: (i) If enough planetary limb was available on the image, the limb was automatically detected and fitted to an ellipse using a least-squares algorithm (Barrey 1984). (ii) In the other cases pairs of NA and WA frames simultaneously

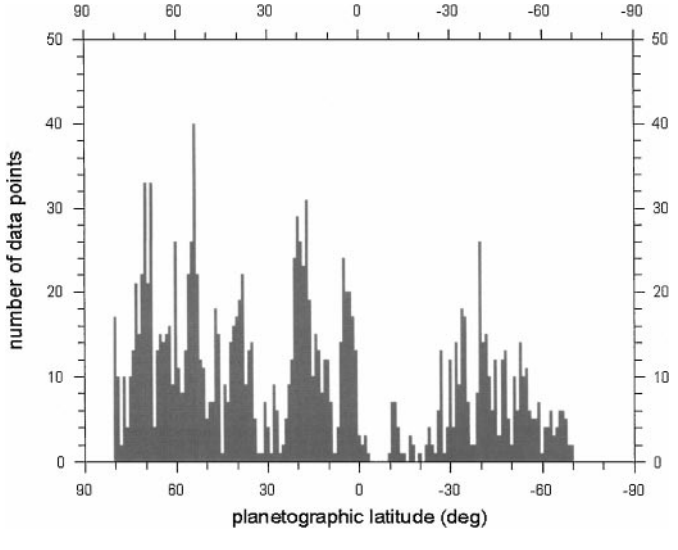


FIG. 1. Distribution of the number of measured wind vectors in 1° latitude bins.

shuttered must be used. The planet center location in the NA image is then determined by fitting an artificial limb to the planet's edge of the corresponding WA image, using a procedure similar to that described in Sada *et al.* (1996). Wind velocities are measured from the calibrated images by comparing the positions of cloud targets at different time-steps. Feature identification and tracking were manually performed on the side-by-side (or blinking) pair of images separated typically by ~10 h (about one planet rotation), although for some pairs we used time separations of 2 and 3 rotations in order to decrease the wind velocity errors.

To improve the quality of the data most measurements were performed with tracers located at distances (in longitude) of no more than $\pm 50^\circ$ from the central meridian, with the exception of the post-encounter *Voyager 2* images used to evaluate the southernmost latitudes. In some cases we also made cylindrical projections and corrected for limb darkening using a standard Minnaert function. These were used as a quality control to constraint the formal errors in the measurements as discussed below.

2.4. Wind Velocity Measurements

The fit of the ellipse to the planetary limb requires precise knowledge of the equatorial and polar radii of the ellipsoid defining the figure of Saturn (R_e and R_p respectively). This fit depends on $(R_e/R_p)^2$ (see, e.g., Barrey 1984). This is also essential when transforming from planetocentric (θ) to planetographic (φ) latitudes:

$$\tan\varphi = \left(\frac{R_e}{R_p}\right)^2 \tan\theta. \tag{1}$$

Once the ellipse is fitted, VICAR transforms the Cartesian coordinates (x, y) of the feature in the image to longitude (λ) and latitude (θ and φ) on the planet disk. The wind measurements are measured relative to the rotating frame defined by the Saturn kilometric radiation period (SKR), which is assumed to be that of the electrically conducting interior, and is known as System III (Desch and Kaiser 1981). However, see Ingersoll *et al.* (1984) and Allison and Stone (1983) for a discussion on the validity of the SKR period as a reference frame. Initially we adopted the SEDR file values for the radius: $R_e = 60{,}330$ km, $R_p = 55{,}045.6$ km. However, Lindal *et al.* (1985) determined different values for the polar radius following a detailed study of Saturn's shape. They obtained $R_e = 60{,}367 \pm 4$ km and $R_p = 54{,}438 \pm 10$ km at pressure level $P = 100$ mbar, but $R_e = 60{,}268 \pm 4$ km and $R_p = 54{,}364 \pm 10$ km at pressure level $P = 1$ bar. Since the cloud tops of most features are assumed to be between these two levels (see Section 5), we have finally adopted the following values of the Saturn spheroid and System III rotation period for latitude positioning and wind velocity measurements:

$$R_e = 60{,}330 \text{ km}$$
$$R_p = 54{,}400 \text{ km},$$

$\tau_{III} = 10$ h, 39 min., 22.4 s ($\Omega_{III} = 810.7939024$ deg/day) as in Davies *et al.* (1986).

From two measurements of the System III longitudes and planetographic latitudes (λ_0, φ_0) and (λ_1, φ_1) of a tracer at times t_0 and t_1, we calculate the meridional (v) and zonal (u) components of velocity according to the following steps. First we transform planetographic to planetocentric latitudes and calculate the meridional displacement $\Delta\theta = \theta_1 - \theta_0$ (in degrees) and average latitude $\theta = (\theta_1 + \theta_2)/2$. Similarly, we obtain the zonal translation $\Delta\lambda = \lambda_1 - \lambda_0$ (in degrees) and time interval $\Delta t = t_1 - t_0$ (in hours). The local radius (in kilometers) at the latitude θ of the feature is given by

$$R(\theta) = \frac{R_e R_p}{\sqrt{R_e^2 \sin^2\theta + R_p^2 \cos^2\theta}}. \tag{2}$$

The meridional velocity is given (in meters per second) by

$$v = R(\theta)\left(\frac{\Delta\theta}{\Delta t}\right)\left(\frac{\pi}{180}\right)\left(\frac{1}{3.6}\right), \tag{3}$$

where $1/3.6$ is the conversion from kilometers per hour to meters per second. The zonal velocity is (in meters per second)

$$u = 2\pi R(\theta)\cos\theta\left(\frac{1}{3.6}\right)\left(\frac{\Delta\lambda}{360\Delta t}\right). \tag{4}$$

2.5. Measurement Errors

As discussed in detail by Ingersoll *et al.* (1981) there are two main sources of errors in the above methodology: (i) image navigation and (ii) target identification. The fitting of an ellipse to the planet's limb (navigation) is the main source of error. It depends on the methodology employed to retrieve the absolute position of the targets (single NA frames or NA + WA simultaneously shuttered frames). Camera pointing performed on narrow angle (NA) frames needs limb arcs as large as possible to improve the fit to the assumed limb model. On the other hand, the location of the limb in the WA frame and the transfer of the derived planet center to the simultaneously shuttered NA frame suffers from seven times lower resolution of the WA frame relative to the NA, reducing the accuracy of the fit. To control this error we have made measurements of the same targets on NA single frames (with small arc limbs of 15°), and in NA + WA pairs, using the same NA frame but fitting the limb with the simultaneously shuttered WA frame. We found that the absolute positioning of the targets depends on their latitude with the following maximum differences in the absolute positioning:

$$\delta\varphi = \varphi(\text{WA} + \text{NA}) - \varphi(\text{NA}) \sim 1^\circ \quad \text{at } \varphi \geq 70^\circ$$
$$\delta\varphi = \varphi(\text{WA} + \text{NA}) - \varphi(\text{NA}) \sim 0.2^\circ \quad \text{at } \varphi < 70^\circ$$
$$\delta\lambda = \lambda(\text{WA} + \text{NA}) - \lambda(\text{NA}) \sim 5^\circ \quad \text{at } \varphi \geq 70^\circ$$
$$\delta\lambda = \lambda(\text{WA} + \text{NA}) - \lambda(\text{NA}) \sim 1^\circ \quad \text{at } \varphi < 70^\circ.$$

In the polar regions the errors are very large and restrict the use of single NA frames for limb fitting to those with limb arcs larger than 45°. With our image selection for the analysis of zonal motions, we estimate that typical navigation errors can contribute to velocity errors $\delta u_n \sim 10\ \mathrm{m\,s^{-1}}$ and to latitude positioning errors of $\delta\varphi_n \leq 0.5°$ (decreasing in the equatorial region).

Target identification and manual pointing of the cursor on the feature are the other sources of error. Saturn cloud features have low contrast and some of them change their shape and brightness, or even disappear, in the typically ~10-h interval between frames. This is particularly important for the peaks of the narrow eastward jets, where the zonal shear destroys features in a characteristic time $\sim 1/(\Delta u/\Delta y) \sim 1/(150\ \mathrm{m\,s^{-1}}/10^6\ \mathrm{m}) \sim 2$ h, and in the active convective areas. In order to control this error, the targets were selected by their brightness, and identification in the two frames was based on the repetitive patterns they formed. Cursor pointing was done on the brightest part of the feature or on its geometric center.

Possible misidentification of the targets could also produce large velocity errors. For example, two features not properly identified and separated by 500 km would give a velocity error $\sigma(u) \sim 500$ km/10 h $\sim 14\ \mathrm{m\,s^{-1}}$. A formal pointing error of 1 pixel (take 100 km) within the selected cloud element will represent an additional error to $\sigma(u)$ of $\sim 3\ \mathrm{m\,s^{-1}}$. Related latitude pointing offsets are $\sigma(\varphi) \sim 0.1°$, but could reach 0.5° or more in the polar areas.

According to these estimations, for a single target, navigation and cloud pointing and identification errors could produce worst-case uncertainties in the resulting wind velocities and latitude locations of

$$\delta u = \left(\delta u_n^2 + \sigma(u)^2\right)^{1/2} \sim 17\ \mathrm{m\,s^{-1}}$$

$$\delta\varphi = \left(\delta\varphi_n^2 + \sigma(\varphi)^2\right)^{1/2} \sim 0.6°.$$

As discussed by Ingersoll *et al.* (1984), these errors, when properly averaged within a latitude box to retrieve the mean zonal wind velocity profile, are not very important since zonal velocities are large in Saturn. For example within a typical 1° box, the statistical error in velocity due to misidentification and pointing errors will be $\sim\sigma(u)/N^{1/2} \sim 3\ \mathrm{m\,s^{-1}}$ (N is the number of points in the box, take $N = 20$). Navigation errors will contribute $\sim\delta u_n/N_n^{1/2} \sim 3\ \mathrm{m\,s^{-1}}$ (N_n is the number of frames used in this box, take $N_n \sim 10$).

However, the determination of the meridional component of the velocity (v) and the eddy components of the motions (u', v') with typical values $<10\ \mathrm{m\,s^{-1}}$ is strongly affected by these errors and casts serious doubts on analysis of eddy momentum transport and energy conversion rates based on this methodology.

3. THE VOYAGER ZONAL WIND PROFILE

3.1. Method of Analysis and Results

The entire set of 2074 zonal wind velocity measurements (*Voyager 1* and *2* data) are plotted vs latitude in Fig. 2A. Independent *Voyager 2* measurements by two of us give very good agreement in general. The scatter in the data found around 30°N and in some eastward and westward jets can be due in part to navigation errors and regional motions, as we will discuss in Section 5. A second important result from this plot is that there is a good agreement between the *Voyager 1* and *Voyager 2* measurements at all latitudes. No apparent short-term temporal changes are noted within the measurement uncertainties.

The next step was to retrieve the averaged zonal wind profile. This average must be understood as a temporal and spatial mean. Temporally, *Voyager 1* and *2* data were separated by ~9 months, with each *Voyager 1* and *2* sampling spanning 1 and 2 weeks around the encounter dates, respectively. Spatially, we have averaged the data in half degree latitude bins. In addition, planetwide longitudinal coverage of the tracers was practically attained at all latitudes except in a few regions (those with few points in Fig. 1).

The zonally averaged wind velocity vs latitude profile was obtained in the following manner. First we filtered the data by removing unrealistic values. Second, we performed a linear least-squares fit of the data points within boxes of 0.5°. Individual measurements with speeds $\pm 25\ \mathrm{m\,s^{-1}}$ from the fit (in each box) were rejected. This represents 16 points of the whole data set. The filtered set (u_i, φ) was the input used to retrieve our smoothed average profile. This was obtained employing a local regression method known as "loess" (Cleveland and Grosse 1991 and references therein). The loess method is a nonparametric procedure for estimating regression functions by fitting parametric functions locally to a scatter plot using a weighted least-squares procedure. The weighting scheme depends on the distance between each point and the point being fitted. In fact, this method represents an extension of the classical local fitting of polynomials. Its strength resides in its ability to effectively manage data points that are not equally spaced, as is our case. The level of smoothing (breadth, or filtering window) is defined by the percentage of data points contained in the window. After several tests, we estimated that 0.3–2.5% levels gave a good representation for the smoothed averaged profile and for the peak values of these jets. Once the loess smoothed profile was retrieved (u_{il}, φ), the final mean zonal profile was obtained by averaging the loess fitted points in 0.5° latitude bins. For a box containing N points u_{il}, the mean zonal wind velocity ($\langle u_\varphi\rangle$ at planetographic latitude φ) is given by

$$\langle u_\varphi\rangle = \frac{\sum_{i=1}^{N} u_{il}}{N} \tag{5}$$

with uncertainty

$$\mathrm{rms}(u_\varphi) = \frac{\sum_1^N (u_i - \langle u_\varphi\rangle)^2}{N-1}. \tag{6}$$

Figure 2B shows the loess smoothed velocity profile compared with the individualy measured data points. The final nominal zonal wind profile for Saturn is numerically presented in Table II.

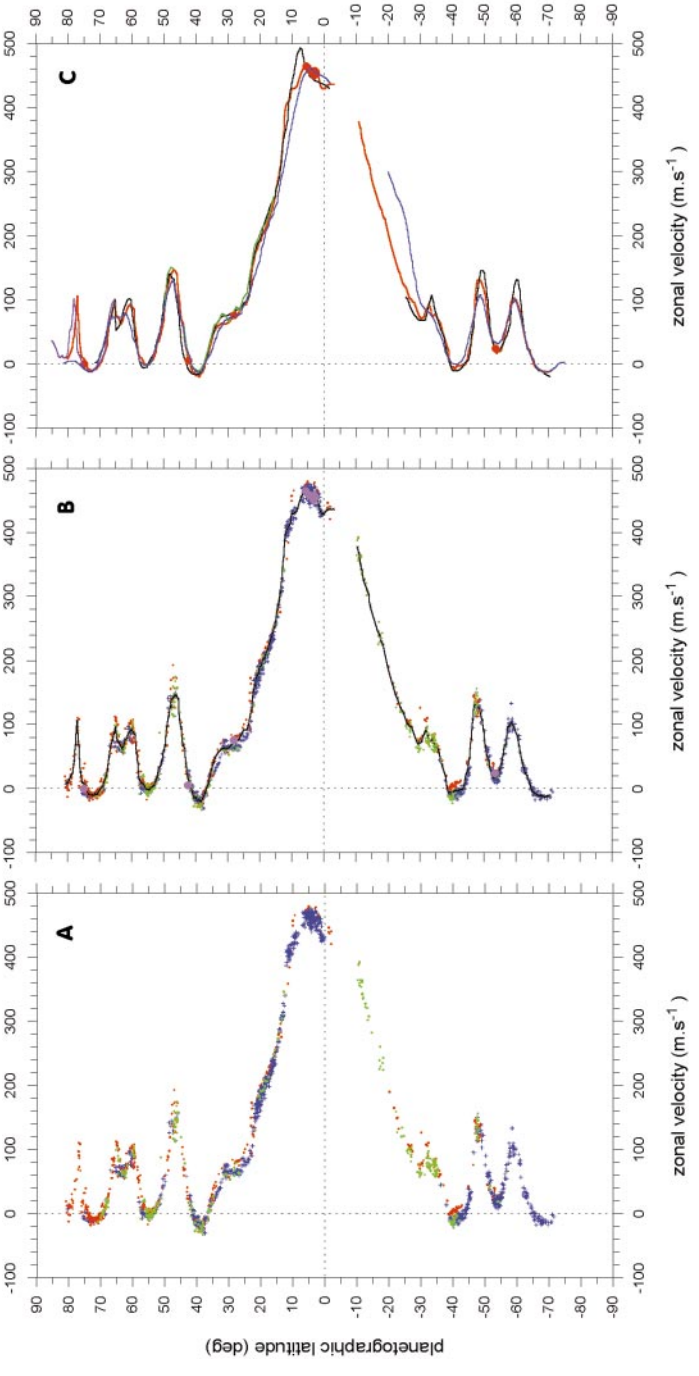


FIG. 2. (A) The 2074 zonal velocity points vs latitude measured in the ISS *Voyager 1* (green dots, Sanchez-Lavega) and *Voyager 2* (red dots, Sanchez-Lavega; blue crosses, Sada) images. (B) The zonal mean velocity as a function of planetographic latitude for Saturn's atmosphere obtained after smoothing, and averaging in 0.5° bins, the 2074 points measured using *Voyager 1* and *2* green images. The large dots correspond to features tracked over a longer time interval and are used as a control for the profile. (C) Comparison between our mean zonal profile (red line) and those of Ingersoll *et al.* (1984, black line), Sromovsky *et al.* (1983, green line), Beebe (1990, in Barnet *et al.* 1992a, blue line), Godfrey (1988, violet line).

TABLE II

Averaged Mean Zonal Flow on Saturn as a Function of Latitude as Determined from Voyager 1 and Voyager 2 Green-Filtered Images

θ (deg)	φ (deg)	<u> (ms^{-1})	rms (u) (ms^{-1})	θ (deg)	φ (deg)	<u> (ms^{-1})	rms (u) (ms^{-1})	θ (deg)	φ (deg)	<u> (ms^{-1})	rms (u) (ms^{-1})	θ (deg)	φ (deg)	<u> (ms^{-1})	rms (u) (ms^{-1})
78.6	80.7	9.9	8.7	42.8	48.7	128.0	12.5	10.9	13.3	327.9	13.0	-31.2	-36.7	55.6	8.3
78.0	80.2	8.3	7.2	42.3	48.2	140.0	12.6	10.5	12.8	362.6	12.0	-31.6	-37.2	46.0	9.7
77.5	79.7	14.1	5.4	41.8	47.7	142.3	13.2	10.0	12.3	396.0	14.2	-32.1	-37.7	37.1	9.7
76.9	79.3	20.4	16.1	41.3	47.2	146.0	19.9	9.6	11.8	405.7	7.8	-33.0	-38.7	20.5	9.7
76.3	78.8	24.8	13.9	40.8	46.7	144.9	11.8	9.2	11.3	409.2	11.1	-33.5	-39.2	2.2	14.3
75.7	78.3	43.5	13.5	40.3	46.2	139.5	11.1	8.8	10.7	415.3	10.7	-34.0	-39.7	-6.3	10.2
75.1	77.8	67.8	15.7	39.8	45.7	107.7	11.1	8.4	10.2	430.3	9.7	-34.5	-40.2	-3.9	10.3
74.5	77.3	105.7	8.2	39.3	45.2	94.9	10.4	7.9	9.7	428.7	9.9	-34.9	-40.7	-9.0	18.2
73.9	76.8	67.2	19.6	38.8	44.7	68.5	9.6	7.5	9.2	430.9	10.2	-35.4	-41.2	-3.8	13.0
73.3	76.3	25.5	12.8	38.3	44.2	62.6	9.0	7.1	8.7	434.0	15.3	-35.9	-41.7	-2.1	12.2
72.7	75.8	9.3	18.2	37.8	43.7	34.3	17.1	5.8	7.2	456.4	19.3	-36.4	-42.2	-3.6	8.6
72.2	75.3	9.5	24.3	37.3	43.2	20.6	15.9	5.4	6.7	461.6	14.3	-36.9	-42.7	-3.0	10.5
71.6	74.8	5.9	9.3	36.9	42.7	11.4	16.5	5.0	6.1	466.0	9.3	-37.3	-43.2	-0.5	16.1
71.0	74.4	-3.6	8.4	36.4	42.2	5.4	13.3	4.6	5.6	467.1	6.5	-37.8	-43.7	-1.9	9.6
70.4	73.9	-8.8	8.0	35.9	41.7	-6.6	13.8	4.2	5.1	465.5	11.0	-38.3	-44.2	0.1	9.1
69.8	73.4	-8.9	6.7	35.4	41.2	-13.9	8.0	3.8	4.6	465.6	20.5	-38.8	-44.7	7.3	5.7
69.3	72.9	-10.1	5.8	34.9	40.7	-12.9	6.1	3.3	4.1	461.0	13.3	-39.3	-45.2	22.4	11.3
68.7	72.4	-10.0	4.9	34.5	40.2	-18.0	13.8	2.9	3.6	459.2	9.2	-39.8	-45.7	28.7	13.5
68.1	71.9	-7.5	5.2	34.0	39.7	-17.3	9.8	2.5	3.1	457.8	21.9	-40.3	-46.2	48.3	7.3
67.5	71.4	-8.5	5.6	33.5	39.2	-20.4	6.6	2.1	2.6	458.0	12.7	-40.8	-46.7	77.9	10.6
67.0	70.9	-5.0	5.9	33.0	38.7	-15.0	12.8	1.7	2.1	458.2	9.9	-41.3	-47.2	108.3	18.6
66.4	70.4	-2.5	7.1	32.6	38.2	-12.1	11.5	1.3	1.5	441.4	17.8	-41.8	-47.7	129.4	7.9
65.8	69.9	2.0	9.1	32.1	37.7	-5.7	8.1	0.8	1.0	431.7	8.0	-42.3	-48.2	131.7	18.3
65.3	69.5	4.5	6.3	31.6	37.2	2.6	12.5	0.4	0.5	429.5	7.4	-42.8	-48.7	129.8	18.7
64.7	69.0	13.5	6.3	31.2	36.7	9.1	6.4	0.0	0.0	429.1	5.7	-43.3	-49.2	125.2	13.5
64.1	68.5	20.6	6.2	30.7	36.1	19.4	11.4	-0.8	-1.0	435.5	5.7	-43.8	-49.7	118.0	9.5
63.6	68.0	28.7	11.4	30.2	35.6	30.3	7.2	-1.3	-1.5	435.9	9.7	-44.3	-50.2	107.6	19.7
63.0	67.5	50.6	12.0	29.8	35.1	39.5	14.4	-1.7	-2.1	436.3	19.7	-44.8	-50.7	89.3	19.7
62.4	67.0	69.1	14.3	29.3	34.6	48.3	6.3					-45.3	-51.2	65.7	17.8
61.9	66.5	76.4	8.7	28.4	33.6	57.7	7.9	-8.8	-10.7	377.1	18.3	-45.8	-51.7	50.0	19.0
61.3	66.0	88.0	12.9	27.5	32.6	61.8	7.9	-9.2	-11.3	368.0	19.9	-46.3	-52.2	43.8	9.8
60.8	65.5	96.4	15.5	27.0	32.1	63.5	10.2	-9.6	-11.8	348.9	14.2	-46.8	-52.7	31.4	9.3
60.2	65.0	75.2	16.8	26.6	31.6	62.9	8.1	-10.0	-12.3	341.1	12.8	-47.3	-53.2	23.5	6.1
59.6	64.5	73.0	9.8	26.1	31.1	62.2	7.5	-10.5	-12.8	325.8	12.9	-47.9	-53.7	16.1	16.1
59.1	64.0	67.6	5.3	25.7	30.6	62.9	9.6	-11.3	-13.8	310.7	18.5	-48.4	-54.1	20.1	5.7
58.5	63.5	61.8	7.7	25.2	30.1	64.5	11.5	-11.7	-14.3	302.9	13.0	-48.9	-54.6	19.0	9.4
58.0	63.1	67.3	7.4	24.8	29.6	67.2	11.5	-12.2	-14.8	286.6	13.0	-49.4	-55.1	22.1	10.4
57.4	62.6	76.1	12.2	23.9	28.6	73.2	12.5	-14.3	-17.4	245.5	13.4	-49.9	-55.6	30.5	9.3
56.9	62.1	79.6	11.4	23.4	28.1	73.3	13.0	-14.7	-17.9	239.8	3.6	-50.5	-56.1	40.5	8.3
56.3	61.6	87.7	10.8	23.0	27.6	76.1	17.4	-15.1	-18.4	229.1	21.7	-51.0	-56.6	53.2	17.5
55.8	61.1	90.8	11.5	22.5	27.0	77.4	17.6	-15.6	-18.9	225.5	19.0	-51.5	-57.1	67.4	18.1
55.3	60.6	93.3	22.6	22.1	26.5	83.6	12.7	-16.9	-20.4	190.7	22.5	-52.0	-57.6	70.4	12.8
54.7	60.1	88.3	9.7	21.2	25.5	92.1	12.3	-18.2	-22.0	164.8	22.5	-52.6	-58.1	95.8	12.7
54.2	59.6	85.2	9.7	20.3	24.5	92.0	14.2	-19.0	-23.0	147.2	12.6	-53.1	-58.6	98.8	7.3
53.6	59.1	56.2	13.3	19.9	24.0	100.7	10.8	-19.5	-23.5	140.7	17.8	-53.6	-59.1	102.8	22.3
53.1	58.6	40.8	16.1	19.5	23.5	112.2	10.3	-20.3	-24.5	125.3	14.0	-54.2	-59.6	99.5	10.0
52.6	58.1	14.7	17.4	19.0	23.0	124.6	12.3	-21.2	-25.5	118.2	7.7	-54.7	-60.1	95.3	15.1
52.0	57.6	8.9	13.3	18.6	22.5	151.1	18.0	-22.1	-26.5	101.8	5.9	-55.8	-61.1	82.1	15.1
51.5	57.1	9.3	13.5	18.2	22.0	162.1	13.2	-22.5	-27.0	100.7	9.2	-56.3	-61.6	66.1	23.0
51.0	56.6	3.9	10.7	17.7	21.5	175.4	14.5	-23.0	-27.6	97.0	10.5	-57.4	-62.6	43.2	5.8
50.5	56.1	-0.3	15.8	17.3	21.0	180.0	7.4	-23.4	-28.1	96.1	10.0	-58.0	-63.1	33.0	6.1
49.9	55.6	-0.8	7.3	16.9	20.4	190.4	11.6	-24.8	-29.6	73.2	8.1	-58.5	-63.5	23.9	4.3
49.4	55.1	-1.9	6.6	16.4	19.9	192.2	10.4	-25.2	-30.1	69.8	9.1	-59.1	-64.0	20.5	4.7
48.9	54.6	1.7	6.3	16.0	19.4	202.3	9.4	-25.7	-30.6	71.7	14.5	-60.2	-65.0	6.1	6.4
48.4	54.1	4.1	5.7	15.6	18.9	204.9	6.0	-26.1	-31.1	73.9	7.3	-60.8	-65.5	-2.0	9.7
47.9	53.7	7.1	5.2	15.1	18.4	212.1	7.9	-26.6	-31.6	82.8	22.8	-61.3	-66.0	-5.6	8.3
47.3	53.2	13.0	6.0	14.7	17.9	217.6	6.7	-27.0	-32.1	88.2	16.4	-61.9	-66.5	-7.2	11.5
46.8	52.7	18.4	5.1	14.3	17.4	226.4	5.9	-27.5	-32.6	94.0	16.5	-62.4	-67.0	-8.9	7.4
46.3	52.2	33.7	4.7	13.9	16.9	234.7	7.0	-27.9	-33.1	84.3	23.8	-63.0	-67.5	-10.3	6.5
45.8	51.7	36.4	5.0	13.4	16.4	243.1	13.4	-28.4	-33.6	74.5	10.3	-63.6	-68.0	-11.1	7.4
45.3	51.2	46.5	5.9	13.0	15.9	252.9	10.2	-28.9	-34.1	74.9	8.3	-64.1	-68.5	-11.5	7.1
44.8	50.7	58.4	10.3	12.6	15.3	261.7	14.9	-29.3	-34.6	77.7	7.7	-64.7	-69.0	-12.1	4.4
44.3	50.2	67.4	16.2	12.2	14.8	278.9	17.5	-29.8	-35.1	71.8	18.0	-65.3	-69.5	-12.9	3.1
43.8	49.7	97.8	20.0	11.7	14.3	292.9	15.4	-30.2	-35.6	66.9	5.5	-65.8	-69.9	-11.9	2.2
43.3	49.2	113.4	18.2	11.3	13.8	307.9	14.2	-30.7	-36.1	63.0	12.8	-66.4	-70.4	-11.4	0.4
												-67.0	-70.9	-9.9	3.6

TABLE III
Control of the Zonal Wind Profile Using Long-Lived Features

Feature	Latitude ($\langle\varphi\rangle$)	$\langle u\rangle$ (feature)	$\langle u\rangle$ [flow at $\langle\varphi\rangle$] (this paper)	$\langle u\rangle$ [(flow at $\langle\varphi\rangle$] (Ingersoll *et al.*)
NPS	75.2 ± 0.8	0.2 ± 0.05	9.5 ± 25	—
BS1	43.1 ± 0.4	5.3 ± 0.1	20.6 ± 16.0	−2
UV Spot	28.7 ± 0.4	76.0 ± 0.1	73.3 ± 12.5	81
EZ-P1	5.6 ± 0.05	464.7 ± 0.6	467.1 ± 6.5	460
EZ-P2	4.6 ± 0.4	458.2 ± 0.6	465.6 ± 20.5	449
EZ-P3	3.7 ± 0.6	454.9 ± 0.6	459.2 ± 9.2	445
EZ-P4	3.2 ± 0.3	451.0 ± 0.6	457.8 ± 22	444
EZ-P5	3.0 ± 0.2	457.8 ± 0.6	457.8 ± 22	443
EZ-P6	2.7 ± 0.3	454.8 ± 0.6	458 ± 12.7	442
Anne	-54.2 ± 0.5	24.0 ± 0.1	20.1 ± 5.7	22

Note. For features identification see Section 3.3. The latitude is given in degrees and the zonal velocities u are given in m s^{-1}.

TABLE IV
Latitudes of Zonal Jet Maxima and Minima and Flow Curvature

Name/comment	$\langle\theta\rangle$ (deg)	$\langle\varphi\rangle$ (deg)	$\langle u\rangle$ (m s^{-1})	$\lvert d^2\langle u\rangle/dy^2\rvert$ (times β)	β ($\times10^{-12}$ m^{-1} s^{-1})
Polar "Hexagon"	74.2	77.3	105.7 ± 8.2	—	1.60
	68.7	72.4	-10 ± 2.7	1.7	2.16
"Double peak"	60.8	65.5	96.4 ± 15.5	16.4	2.88
"Double peak"	58.5	63.5	61.8 ± 7.7	6.0	3.06
"Double peak"	55.3	60.6	93.3 ± 22.6	4.8	3.33
	49.4	55.1	-1.9 ± 6.6	1.6	3.76
"Ribbon" wave	41.3	47.2	146 ± 19.9	5.4	4.31
	33.5	39.2	-20.4 ± 6.6	1.3	4.66
Equatorial jet	3.8	4.6	465.6 ± 20.5	0.6	5.42
	−25.2	−30.1	69.8 ± 9.1	1.4	5.00
Partial wave	−27.5	−32.6	94.0 ± 16.5	1.9	4.93
	−34.9	−40.7	-9 ± 18.2	1.0	4.57
Partial wave	−42.3	−48.2	131.7 ± 18.3	6.9	4.19
	−48.4	−54.1	20.1 ± 5.7	2.9	3.80
	−53.6	−59.1	102.8 ± 22.3	4.9	3.45
	−65.3	−69.5	-12.9 ± 3.1	0.6	2.53

The goodness of the averaging procedure was tested by comparing the retrieved velocities with those of discrete "long-lived" features tracked over a large temporal interval. All tracers used for this purpose exhibited a constant linear longitude drift, so their zonal velocity was derived from a least-squares fit to this trend. Although more data about these features will be presented in Section 3.3, here they are used to control the profile (the drift rates for some particular features are drawn in Fig. 7). We are assuming that they act as passive tracers of the zonal flow. As shown in Table III, there is a good agreement between the individual measurements of the features and the zonal mean flow. The only exception is the Brown Spot 1 (feature BS1), whose velocity differs by ~5 m s^{-1} compared with the mean flow. The reason could be that BS1 has its own velocity component relative to the mean flow, as occurs with some large-scale jovian vortices. Note also that BS1 is embedded in a region with turbulent motions and probable regional velocity components (there is a significant difference between our nominal profile and that of Ingersoll *et al.* (1984) at these latitudes).

In Table IV we present the latitudinal locations of the zonal jet maxima (eastward currents) and minima (westward currents) resulting from our analysis. The nominal profile is then used to retrieve the ambient vorticity (du/dy) and curvature flow profiles (d^2u/dy^2)

$$\frac{d\langle u_\varphi\rangle}{dy} = \frac{\langle u_{\varphi+0.5}\rangle - \langle u_{\varphi-0.5}\rangle}{\Delta y} \tag{7}$$

$$\frac{d^2\langle u_\varphi\rangle}{dy^2} = \frac{\langle u_{\varphi+1}\rangle + \langle u_{\varphi-1}\rangle - 2\langle u_\varphi\rangle}{\Delta y^2} \tag{8}$$

with $\Delta y = (\pi/180)R(\varphi)$. Because of the resulting noisy structure obtained with Eq. (8), the curvature was also calculated by means of a third degree polynomial fit to the jets. Figure 3 shows the ambient vorticity profile and Table IV also gives the maximum value of the flow curvature for the jet peaks. These

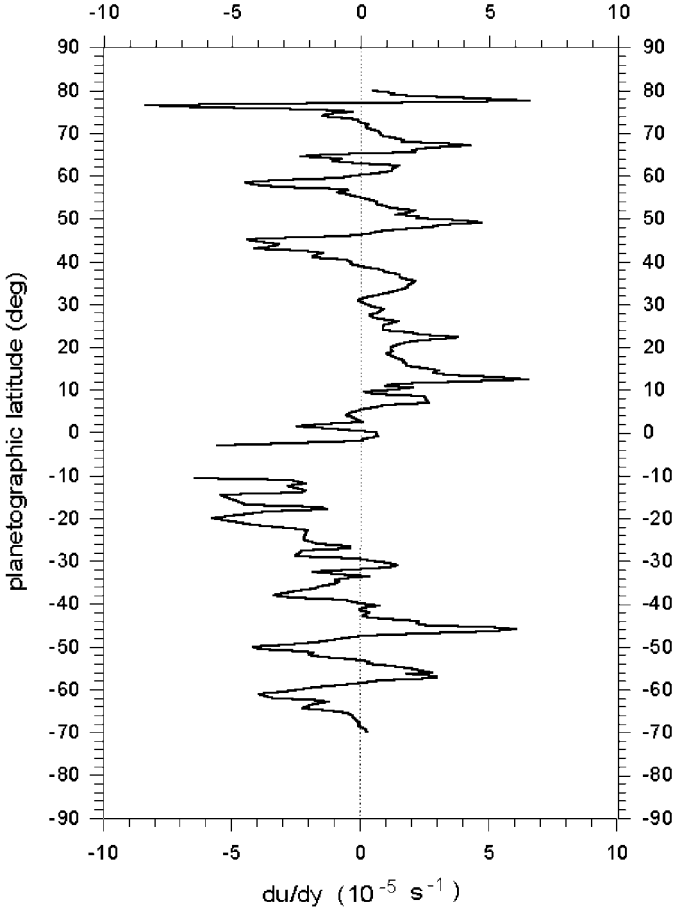


FIG. 3. Meridional profile of the ambient vorticity du/dy (in s^{-1}) of Saturn's zonal flow.

values are compared with the meridional gradient of the ambient vorticity given by

$$\beta = \frac{2\Omega \cos\varphi}{R(\varphi)}. \quad (9)$$

3.2. Comparison with Other Voyager Measurements

We have compared our profile listed in Table II and Fig. 2B with those previously published by Ingersoll *et al.* (1984), Beebe (1990, cited in Barnet *et al.* 1992a), Sromovsky *et al.* (1983), and Godfrey (1988) (see Table I). We take the data from Ingersoll *et al.* instead of Smith *et al.* (1982) since it represents a reworked version according to the authors. Caution must be taken in using the same values of the equatorial and polar radius of the planet for the derived wind velocities when performing these comparisons. Although there is a general agreement among the authors for using the same equatorial radius (60,330 km), there are differences with the polar radius employed. Ingersoll *et al.* (1984) used $R_p = 54{,}450$ km, in accordance with the value suggested by the Image Proccesing Laboratory at JPL (G. Yagi, JPL Interoffice Memorandum 384-IPL-SIPG-83-112, 1983). This value is similar to what we used (see Section 2.1). Beebe used $R_p = 55{,}045$ km, in agreement with the SEDR files and the VICAR manual. Sromovsky *et al.* (1983), on the other hand, used $R_p = 54{,}880$ km. Unfortunately Godfrey (1988) does not give the value of the polar radius he used. Maximum latitude differences for a feature could be as large as 0.66° at 45° planetographic latitude for these extreme values.

The various profiles are compared in Fig. 2C. Except for the case of Sromovsky *et al.*, whose data were presented in a table, the other profiles were retrieved from a scan of the published graphs. While there is good agreement in the general behavior of these profiles (location of the eastward/westward jet peaks, shape and wing structure of the jets), there are also substantial differences that merit some comments. (1) In the northern hemisphere our profile shows a slight latitude offset of ∼0.5° (southward) relative to Ingersoll *et al.*, but agrees very well with that of Beebe. (2) Beebe's profile between −30° and −10° shows a large southward shift that it is not consistent with our data, and with the tendency shown by the Ingersoll *et al.* profile. (3) Our profile, and that of Beebe, does not show the narrow jet peak seen at ∼7°N by Ingersoll *et al.* that is superimposed to the broad equatorial jet. (4) The jet at ∼77°N embedded in the hexagon is distinct in our data. However its position is found to be displaced ∼1° southward relative to that given by Godfrey. (5) We confirm the existence of a tiny peak at latitude −32.5° seen in Ingersoll *et al.*'s data, but not in Beebe's profile. (6) The velocity in the peak of the southern jet at −48° is midway between that of Beebe and Ingersoll *et al.*, and the peak of the −59° jet is more coincident with that of Beebe. These differences among authors must result, apart from the value of the radius used, from the measurement errors listed in Section 2.5, in particular navigation and methodology used with the frame management (single NA or pairs NA + WA) and from data sampling in Beebe's data.

3.3. Complementary Studies of the Morphology and Motions of Particular Cloud Systems

Inspection of the *Voyager 1* and *2* images showed some peculiar features that were not included or described in detail in previous *Voyager* reports. Here we present new measurements and discuss the morphology of these features.

(a) Equatorial zonal plumes. During the *Voyager 2* encounter, the northern EZ showed a family of 5 to 7 active bright spots centered between ∼2° and 6°N that were distributed regularly over about 60° of longitude (Fig. 4). These features were not present during the *Voyager 1* encounter. They were located close to the equatorial jet maximum moving with velocities $u \sim 450$ m s^{-1} (identified as features EZ-P1 to EZ-P6 in Table III). They had a plume-like aspect and were detached as observed in green and orange wavelengths, but were barely visible in the violet filter images. Their morphology is consistent with a rapid vertical convective origin, followed by an east–west spreading of cloud material and equatorward extension due to the meridional shear of the zonal winds ($du/dy \sim 7.9 \times 10^{-6}$). They could represent smaller versions of the rare large-scale events known as "Great White Spots" (Sanchez-Lavega *et al.* 1991).

(b) Southern waves. One surprising result of the Voyager images was the presence of a distinct and unique wave structure at mid-northern latitudes nicknamed "the ribbon" (Smith *et al.* 1981). Careful inspection of Voyager images of the southern hemisphere reveals the presence of a symmetric wave structure at −48° latitude (Fig. 5). This wave is not as prominent as its northern counterpart, but detaches in green-filtered images in some longitude sectors as a "white ribbon." However, blue images showed it as an undulating border separating a lower albedo northern region from a higher albedo southern area. This color behavior is similar to that seen on the northern ribbon (Godfrey and Moore 1986). Another undulating diffuse wave covering short longitude sectors was seen in green images at latitude −32° (Fig. 5). This wave was bright in *Voyager 1* images but dark in *Voyager 2* images, although it was not evident over most longitudes at this latitude. The properties of these waves are given in Table V. Both waves showed similar morphological and spatial properties: wavelength (∼5°–6°), meridional amplitude (∼1°), width (∼0.4°–1.3°), and location in the eastward jet peaks. The measurement of the drift rates of the crests of these waves, when compared to those of individual cloud elements in the same latitude but at different longitudes (in places where the waves are not present), tentatively suggest a phase speed motion relative to the background flow. From Table V we find $\langle c \rangle - \langle u \rangle \sim -15$ m s^{-1}. This aspect is important when establishing the zonal profile, as will be discussed in Section 5.

It is striking to see how well the properties of these waves resemble those of the northern ribbon (Sromovsky *et al.* 1983).

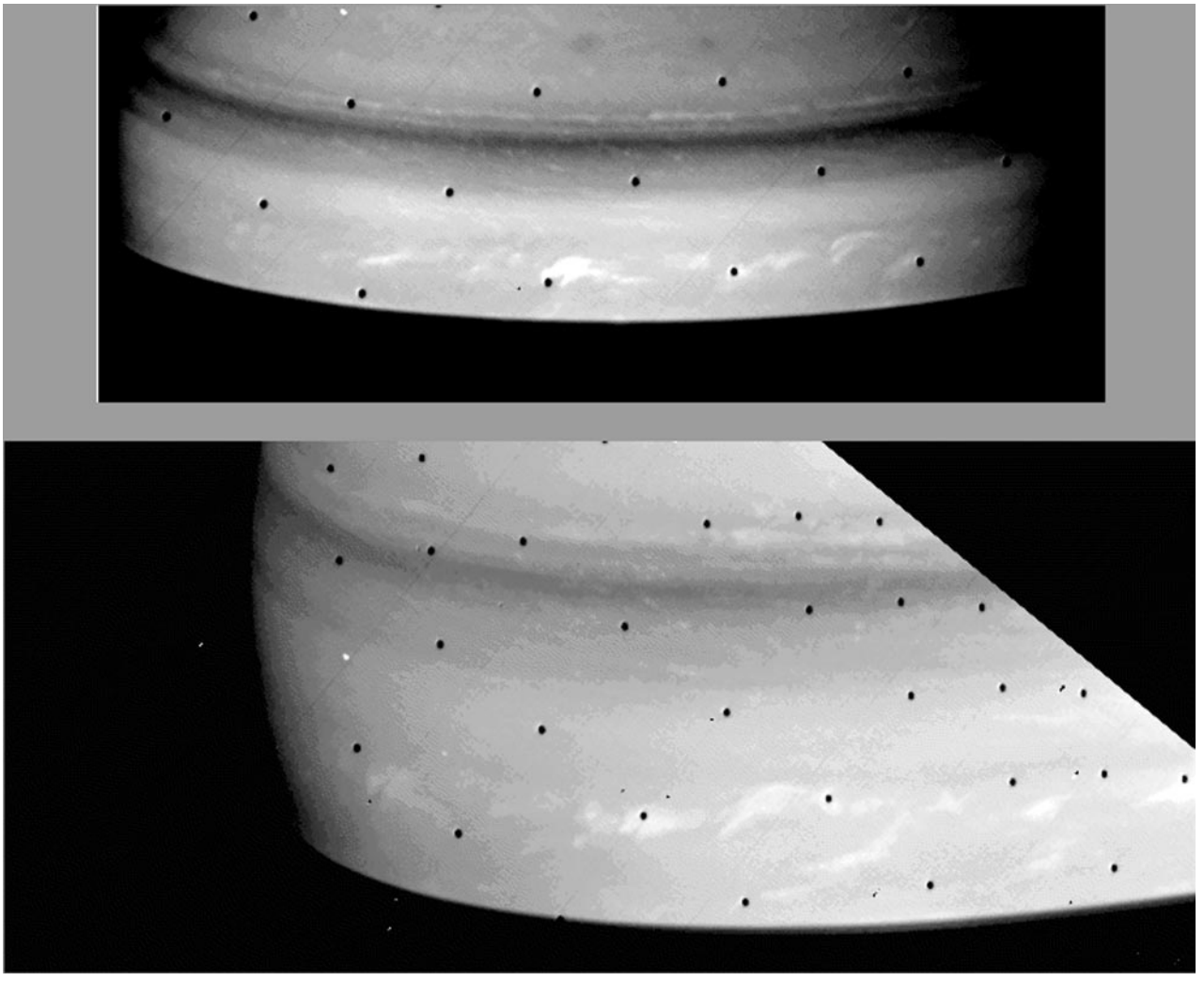


FIG. 4. Aspect of the series of equatorial zone plumes in *Voyager 2* green-filtered images: August 23 (top) and 25 (bottom), 1981.

We are tempted to conclude that a similar instability mechanism (perhaps baroclinic, as suggested by Godfrey and Moore 1986) operates on mid-latitude eastward jets to generate these features. Their strength could be related to the season on Saturn (i.e., to insolation): strong at northern latitudes and weak at southern ones during the *Voyager* encounters.

(c) Long-lived ovals. We have studied the motions and measured the sizes of the largest ovals (diameter > 1000 km) identified in *Voyager 1* and *2* images by Smith *et al.* (1981, 1982). We have found that only four features survived the period between encounters: the North Polar Spot (NPS or "Big Bertha") at 75.2° latitude, the Brown Spot 1 (BS1) at 43.1°, the UV-Spot

TABLE V
Properties of the Wavy Sectors in the Southern Hemisphere

Wave	Date/spacecraft	$\langle\varphi\rangle$	$\langle c\rangle$ (m/s)	$\langle u\rangle$ (m/s)	$\Delta\lambda$	$A(\varphi)$	$\Delta\varphi$
−32°	6–12 Nov 1980 Voyager 1	$-33.3^\circ \pm 0.5^\circ$	73 ± 10	$87 \pm 5^\circ$	$6^\circ \pm 0.4^\circ$	$1.1^\circ \pm 0.3^\circ$	$0.7^\circ \pm 0.3^\circ$
−32°	20–24 Aug 1981 Voyager 2	$-32.6^\circ \pm 0.5^\circ$	—	99 ± 8	$6^\circ \pm 0.4^\circ$	$1.2^\circ \pm 0.3^\circ$	$0.4^\circ \pm 0.2^\circ$
−48°	6–12 Nov 1980 Voyager 1	$-48.8 \pm 0.6^\circ$	133 ± 13	—	$4.7^\circ \pm 1.2^\circ$	$1^\circ \pm 0.5^\circ$	$0.7^\circ \pm 0.2^\circ$
−48°	20 Aug–4 Sept 1981/Voyager 2	$-48.4^\circ \pm 0.6^\circ$	127 ± 10	138 ± 7	$5.8^\circ \pm 0.3^\circ$	$1.3^\circ \pm 0.5^\circ$	$1.3^\circ \pm 0.5^\circ$

Note. The columns are as follows: wave identification, dates, $\langle\varphi\rangle$ = average wave latitude, $\langle c\rangle$ = phase velocity, $\langle u\rangle$ = mean flow speed (see text), $\Delta\lambda$ = wavelength (crests separation), $A(\varphi)$ = meridional amplitude (peak to peak), $\Delta\varphi$ = "channel" width.

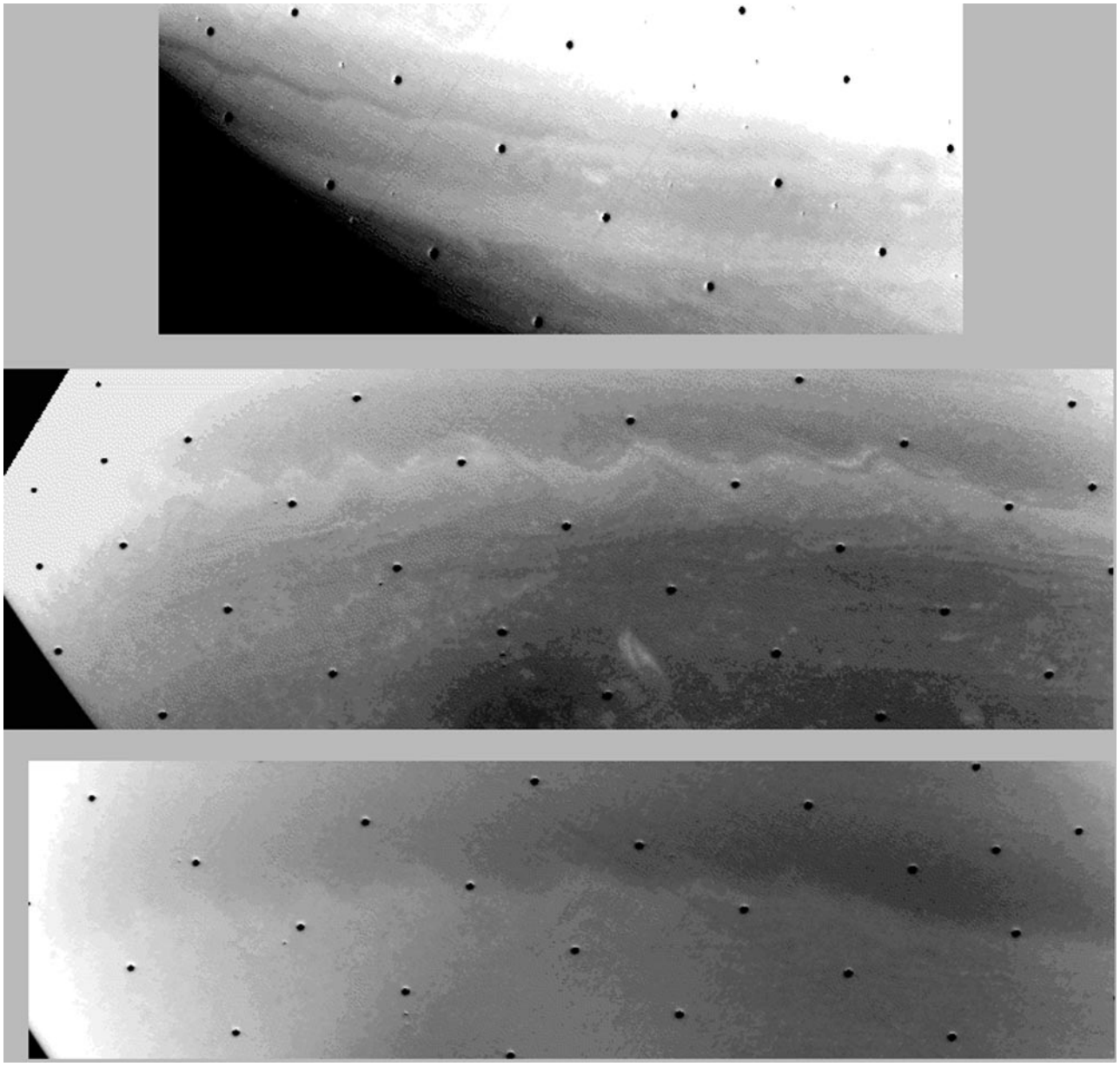


FIG. 5. Aspect of the "ribbon-like" partial waves in the southern hemisphere. The wave at -32° is seen in green wavelengths close to the limb on August 20, 1981 (*Voyager 2*, upper). The wave at -48° is seen in green (middle) and blue (lower) images on November 12, 1980 (*Voyager 1*).

at 28.7°, and the "Anne" Spot at -54.2° (Fig. 6). The NPS was also observed during the 1990s using ground-based telescopes (Sanchez-Lavega *et al.* 1993b, 1997) and HST (Caldwell *et al.* 1993). Additional details are given in these references. The System III longitude drift of the other three features is shown in Fig. 7, and the derived zonal velocities that were used to control the wind profile are given in Table III. Faced with the possibility that some of them could survive longer time periods, and be present during the *Cassini* spacecraft mission, we present in Table VI their motion equation (ephemeris) derived from the drift rates shown in Fig. 7. Precise oval dimensions obtained from a large series of careful measurements are also presented. Note that we have defined the limits of the ovals as the edges of their maximum albedo contrast relative to their surroundings. However, at high resolution, the circulation patterns around the ovals seem to extend further than this edge (Fig. 6).

4. THE GROUND-BASED AND HST ZONAL WINDS

4.1. Ground-based and HST Data Sets

Two groups of telescopic data sets were considered to study the long-term behavior of the zonal winds: (i) Historical (up

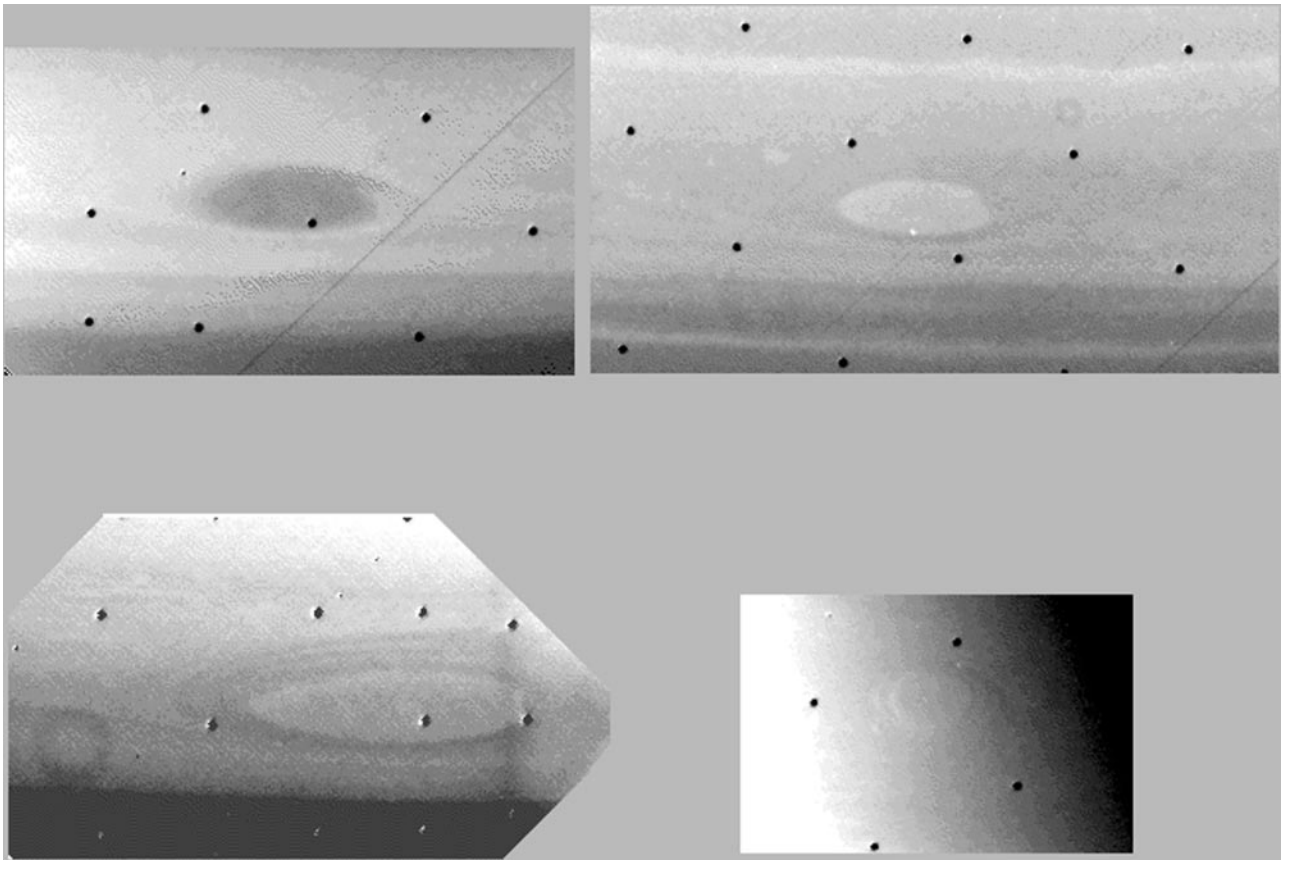


FIG. 6. Different aspects of the long-lived red oval nicknamed "Anne Spot" at -54° latitude. Upper images by *Voyager 1* on November 11, 1980, violet filter (left) and green filter (right). Lower images by *Voyager 2* on August 23, 1981 (left, green filter) and September 4, 1981 (right, green filter).

to 1990), and (ii) contemporary (1990s). Historical data were selected by the reliability of the observations of spots used as tracers for mass motions and originate from various sources (Sanchez-Lavega 1982, Sanchez-Lavega *et al.* 1993a). Ten features observed visually and photographically between 1793 and 1978 were selected. The "contemporary" data comprises two sets obtained following the 1990 Great White Spot eruption (Sanchez-Lavega *et al.* 1991, Beebe *et al.* 1992). One set consists of CCD ground-based telescopic images in red wavelengths and covers the period 1990–1998. A total of 86 wind velocity data points were considered, most of them published in Sanchez-Lavega *et al.* (1993a, 1996, 1999). Other unpublished data from 1992 and 1998 were incorporated into this work. Typical latitude errors in these measurements are $\sim 1^\circ$–2°, with velocity errors ranging from ± 0.5 up to ± 15 m s^{-1}. A large part of these data points (historical and 1990s ground-based) correspond to the equatorial region between 33° and -19° (86 points), with the rest (10 points) located at latitudes northward of 47° and southward of -53°.

TABLE VI
Ephemeris for the Largest and Long-Lived Saturn Ovals

Feature	Size (E–W × N–S)	Motion equation[a]
NPS	11,000 × 3,700 km	$\lambda = 343.15^\circ - 0.0792462\,T - 5.54345 \times 10^{-6}\,T^2$ (1)
BS1	5,000 × 3,800 km	$\lambda = 292.16^\circ - 0.59979\,T$ (2)
UV-Spot	6,000 × 4,900 km	$\lambda = 2376.1^\circ - 7.09475\,T$ (3)
Anne	3,200 × 2,500 km	$\lambda = 1322.35^\circ - 3.64853\,T$ (4)

[a] λ is System III longitude, and the ephemeris time T represents the days since 27 Sept. 1980 (12:00 UT) or Julian Date = 2,444,510.0. (1) The NPS observing period has limiting observing dates from 29 Sept. 1980 to 23 Aug. 1981 (*Voyager 1* and *2*) complemented with ground-based data (7 Apr. 1990 to 6 Nov. 1995); see Sanchez-Lavega *et al.* (1993b, 1997). (2) BS1 was measured from limiting dates 27 Sept. 80 to 24 Aug. 1981. (3) UV-Spot was from 9 Oct. 1980 to 24 Aug. 1981, and (4) Anne from 27 Sept. 1980 to 4 Sept. 1981.

The HST data set was obtained in 1990, during the development of the Great White Spots (GWS), by Barnet *et al*. (1992b). Wind velocities were determined between 32° and -9° using a cross-correlation method applied to east–west scans at different latitudes in green (547-nm) and methane band (889-nm) filtered images. In this work we directly take their tabulated latitude vs zonal mean velocities for comparison with our data. A large dispersion is inherent to this measurement technique,

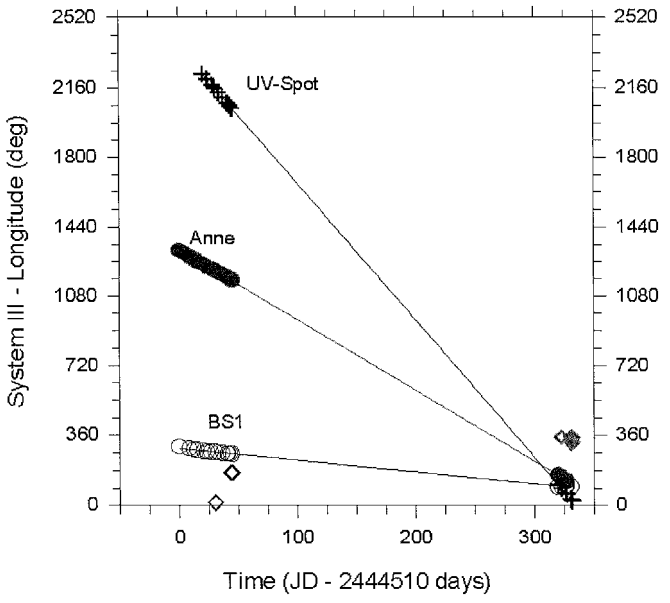


FIG. 7. Longitudinal linear drift rate (in System III) of three long-lived oval spots (UV-Spot, BS1, and Anne) during the *Voyager 1* (left points) and *Voyager 2* (right points) encounters. Other data points correspond to smaller brown spots near the latitude of BS1.

so deviations from the mean as large as $\pm 50\ \mathrm{m\,s^{-1}}$ are expected.

4.2. Comparison of Ground-Based and HST Winds with the Voyagers Profile

In Fig. 8 we compare the *Voyager* profile with the rest of the data available in the equatorial region. We have marked with large dots the cores of the events known as GWSs that occurred in 1876, 1903, 1933, 1990, and the large spot observed in 1994. Important velocity differences are observed between the nominal profile and these data. The most significant occurring at the peak of the equatorial current (between latitudes $\sim +12^\circ$ and -12°), and affecting the individual tracers and the cores of the GWSs. At green–red continuum wavelengths there are velocity differences $\Delta u \sim -50\ \mathrm{m\,s^{-1}}$ between the *Voyager* and the other data, although these differences are larger $\Delta u \sim -120\ \mathrm{m\,s^{-1}}$ at -12°. It is evident that the cores of the GWSs have lower velocities than those obtained by tracking *Voyager* features at the same latitudes. The most significant case is that of the 1994 storm, where the differences found are as large as $\sim -50\ \mathrm{m\,s^{-1}}$ (Sanchez-Lavega *et al.* 1996). Another relevant result found by Barnet *et al.* (1992b) was the lower velocities measured in the 890-nm methane band images. In the next section we will discuss the possible nature of these differences.

Figure 9 compares the few points measured outside the equatorial current with the nominal *Voyager* profile. It seems that, in general, the scarce data points available match the *Voyager* profile within the measurement uncertainties. We are tempted to suggest that the mid-latitude jets could be stable in time on a

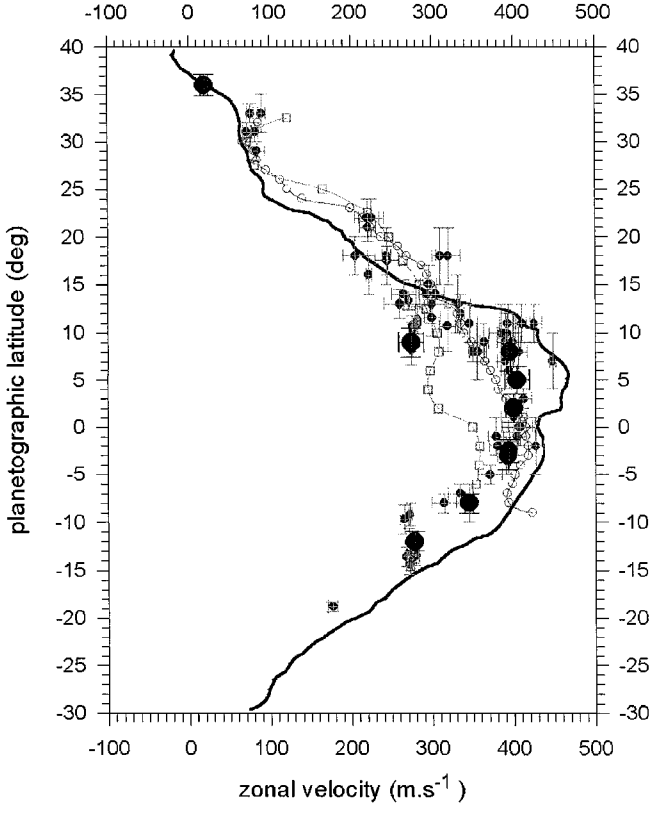


FIG. 8. Saturn's equatorial flow in time. Comparison of the nominal *Voyager* profile (dark line) with the zonal velocities of the GWSs cores (large dots), individual data points measured using ground-based facilities (points with error bars), and with the mean HST profile at 546 nm (circles, thin line) and at 890 nm (squares, thin line).

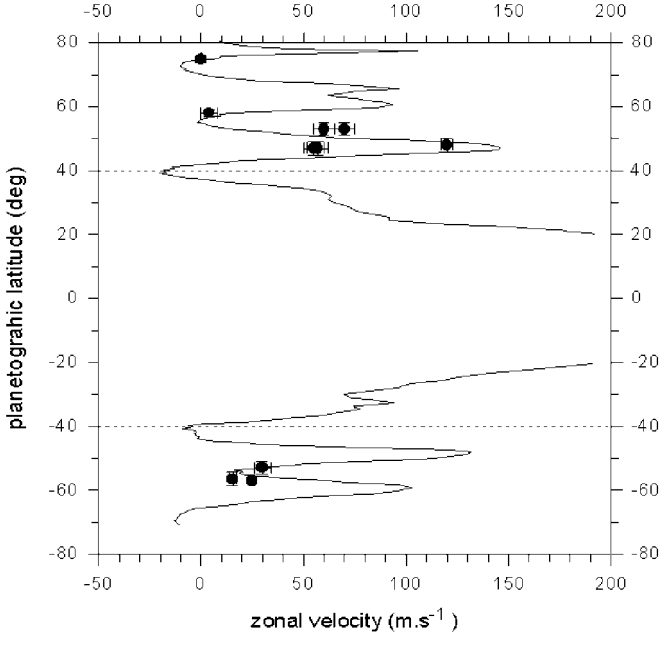


FIG. 9. Saturn's nonequatorial flow in time. Comparison of the nominal *Voyager* profile (continuous line) with the few data points available from ground-based observations (dots).

global scale, but a firm conclusion could not be reached because of the small number of points available.

5. DISCUSSION ON THE PROCESSES INFLUENCING THE ZONAL WIND VELOCITY PROFILE

Aside from the measurement uncertainties, the scatter in the velocities we found at various latitudes could have several origins, including altitude differences of the tracers, particular motions, and structural temporal changes in the global circulation. We discuss these aspects in this section.

5.1. Vertical Shear of the Zonal Wind

A basic aspect when comparing wind profiles is to determine the effective altitude of the cloud tracers. A dependence of the zonal wind velocity on altitude together with tracers placed at different levels will produce different profiles.

Using the Voyager IRIS meridional temperature measurements at various levels, Conrath and Pirraglia (1983) applied the thermal wind relation for geostrophic conditions to determine that the winds decrease globally with height above the clouds. Typical rates are $(\partial u/\partial z) \sim -20$ m s^{-1} per scale height ($H \sim 40$ km at $P \sim 500$ mbar). The location of the cloud tops for the tracers employed in the *Voyager* measurements is not exactly known. Cloud tracers are expected to be located somewhere between ~0.5 and 1 bar ($\Delta z \sim 30$ km) outside of stormy epochs (Tomasko *et al.* 1984). Most Voyager wind measurements were performed at a wavelength of 566 nm (green filter) that senses altitudes deeper than 1 bar in the absence of particles (i.e., if only gas absorption and Rayleigh scattering were present). Thus if tracers are placed at different altitudes within this range a dispersion is zonal wind velocity should be ~15 m s^{-1}.

However, Barnet *et al.* (1992b) measured the motions in the 1990 GWS clouds at two wavelengths that sense different atmospheric levels: 890 nm (methane band) and 547 nm (continuum). As indicated above, they found important differences between the two profiles in the equatorial region, which they propose are due to altitude differences (see Fig. 8). According to a recent radiative transfer model that helps interpret the altitude of the GWS storm clouds (Acarreta and Sanchez-Lavega 1999), cloud tops were located at $P \sim 50$–100 mbar (890 nm) and $P \sim 200$–300 mbar (547 nm), representing an altitude difference $\Delta z \sim 40$–60 km. Using the Barnet *et al.* averaged wind data at these two wavelengths, a maximum vertical wind shear of $(\partial u/\partial z) \sim -70$ to -90 m s^{-1} per scale height is found at latitude $+5^\circ$.

Our observations of the 1990–1994 equatorial storms in the methane bands filters confirm their high altitude (Sanchez-Lavega *et al.* 1994, 1996). In particular, the 1994 storm was detected in the deep methane band at 2.3 μm, indicating that its altitude reached at least the ~50-mbar level. The same could be assumed to be valid for the other long-lived spots plotted in Fig. 8. Their motions differ markedly from the *Voyager* wind profile, so a high altitude location of their clouds is the most plausible explanation for the different velocities found for large storm cores.

5.2. Regional Motions

The cloud elements used as tracers for the zonal flow have been assumed to be "passive," i.e., inert. However, the morphology of the cloud systems at some latitudes on Saturn clearly indicates, as discussed above, the presence of waves and "active" (rapidly changing) cloud forms. Wind data derived from tracking these clouds could mask the real zonal flow and introduce scatter in the wind velocity profile.

(a) Waves: Visually undulating cloud patterns. Tracking of the crests and troughs of wave structures (such as the ribbon at 45°, or the partial waves in the southern hemisphere at -32° and -48°) yield their Doppler-shifted phase velocity $c - \langle u \rangle$. Atmospheric waves show a dispersion relation $c(k)$, with k being the wavenumber, that is given by the nature of the wave. Direct measurements of these parameters were performed by Sromovsky *et al.* (1983) for the 45° ribbon. They found $c \sim 70$ m s^{-1} for $k = 5$ (wavelength 72°), but $c \sim 140$ m s^{-1} for $k = 36$–51 (wavelengths ~ 10°–7°). However, the latter represent the most visually distinctive wave components of the ribbon. These features moved very close to the zonal mean velocity of the flow $\langle u \rangle \sim 145$ m s^{-1} according to our tracking and that of Sromovsky *et al*. For the southern waves at -32° and -48° we found $c - \langle u \rangle \sim 15$ m s^{-1} (see Table V). The measurements of the zonal winds in regions where the waves are present could show higher scatter if tracers are not properly identified and separated from the wave pattern.

(b) Wave–spot longitude oscillations. Oscillations in longitude of the tracers, as those measured in a long-lived spot seen during 1969 and 1970 (Reese 1971), could introduce some scatter in $\langle u \rangle$. However, we must say that such oscillations have not been observed for the major features tracked during the *Voyager* encounters (see Fig. 7).

(c) Wave motion related to the 1990 GWS. The 1990 GWS showed regularly spaced cloud patterns in longitude, reminiscent of a wave phenomena, during its development stage (Sanchez-Lavega *et al.* 1991, Westphal *et al.* 1992). Indeed, Barnet *et al.* (1992b) detected a zonal dependence in the wind velocity that could have been due to forced waves. They noted a strong decrease of the wind speeds eastward of the storm nucleus (rate of 1.2×10^{-3} m s^{-1} km^{-1}). For two storm features separated in longitude by ~90° we get a velocity difference $\Delta u \sim 110$ m s^{-1}, a large value that could be responsible in part for the dispersion found in Fig. 8.

(d) Active cloud systems. Local motions related to extended and active cloud systems can introduce scatter in the zonally steady wind flow measurements. Examples are as follows: (i) Spots exhibit divergence as the V-shaped convective features seen at 42° (Hunt *et al.* 1982) and the core of the 1990 equatorial storm during its growing stage (Sanchez-Lavega 1982, Sanchez-Lavega *et al.* 1991, Beebe *et al.* 1992). If L is the scale of the

motions and the measured diverge field is div V, from the values given by these authors, we get $u' = L \text{ div } V - \langle u \rangle \sim 50 \text{ m s}^{-1}$. (ii) There can be circulation around closed vortices. For example, Smith *et al.* (1982) measured anticyclonic circumferential flow speeds of $u' = 30 \text{ m s}^{-1}$ in feature BS1. The UV-spot at $+28.7^\circ$ showed an extended cyclonic circulation pattern of clouds around the core (*Voyager 2* epoch), but care was taken not to include features from this pattern in the wind profile. (iii) Complex eddy motions (turbulence, spot orbiting, vortex streets) have been studied by Sromovsky *et al.* (1983) and by Godfrey *et al.* (1983) for latitudes $+29^\circ$ to $+46^\circ$. Typical velocities attributed to these eddies are $u' \sim 10$–15 m s^{-1}.

5.3. Structural Changes in Saturn's Circulation

The velocity differences seen between the *Voyager* wind measurements and the ground-based and HST data sets in the equatorial jet could also be due to structural changes in the jet itself. This can occur, for example, if there is a meridional momentum transfer (and kinetic energy) to the mean flow by the eddy motions present ($d\langle u'v' \rangle / dy$), and by the mean flow itself ($d\langle u \rangle \langle v \rangle / dy$). This aspect has been investigated in Saturn's northern hemisphere by Ingersoll *et al.* (1984) and by Sromovsky *et al.* (1985), although no conclusive result has been achieved due to the large uncertainties of the retrieved meridional velocities. However, momentum transfer can be very effective during the development of large-scale disturbances as the GWS, in particular if coupled to the vertical momentum transport ($d\langle u'w' \rangle / dz$) of the suspected waves triggered during the initial development of the disturbance.

On the other hand, the zonal circulation in the upper troposphere can be controlled (or modulated) in part by the seasonal insolation cycle, which is important on Saturn due to the planet's oblateness and its rotational axis tilt. Ring shadowing increases this effect over equatorial latitudes. Conrath and Pirraglia (1983) showed the existence of a seasonal modulation effect on Saturn's temperature field in the lower stratosphere and upper troposphere. A radiative-dynamical model by Barnet *et al.* (1992a) suggests that winds driven by insolation changes can reach amplitude maximums in velocity of $\sim$10–20 m s^{-1} at pressure levels $P \sim 80$–110 mbar between opposite seasons. These effects seem insufficient to explain the large differences seen in Fig. 8.

6. CONCLUSIONS

A reanalysis of *Voyager 1* and *2* images obtained in 1980 and 1981 and a compilation of historical and modern ground-based and HST observations have been used to study the zonal wind velocity profile in Saturn's atmosphere at cloud level. A detailed analysis of the data, including independent measurements and careful error control, has allowed us to present a nominal profile for the latitude range $+81^\circ$ to -71° that can be used as a reference for the forthcoming exploration of the planet by the *Cassini* spacecraft. The *Voyager* images have also been used to study some cloud systems in the equator and south hemisphere of the planet not previously described in the literature. A comparison of the *Voyager* nominal profile with historical and recent data shows significant differences in the equatorial jet, but not at other latitudes. These differences could be due to the different altitudes of the clouds used as tracers coupled with vertical wind shear, particular regional motions (waves, convection, turbulence), and even in part to a seasonal structural change in Saturn's equatorial circulation. It seems most plausible that features pertaining to large-scale disturbances, like the Great White Spots, are not passive tracers of the mean flow and that the disturbance itself could have modified the jet profile. Seasonal insolation and ring-shadowing effects on circulation could also be involved in the temporal differences found for both sides of the equator. At present we can only partially constrain and separate each of these effects on the measured differences.

A primary task of the *Cassini* mission is to study the general circulation and meteorology of Saturn's atmosphere. One objective should be the determination of the altitude of the targets used as tracers of the winds, and the clear separation of regional motions from the general zonal flow. This mission will take place about one saturnian year after the *Voyager* missions. In between, a large storm took place in the equator (in 1990), with medium and minor scale disturbances occurring in subsequent years (1991, 1994–1998). It would be very interesting to see how this has affected the atmospheric circulation of the equatorial latitudes, and to use this information to improve our knowledge of the mechanisms that drive the strong zonal winds on the planet.

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